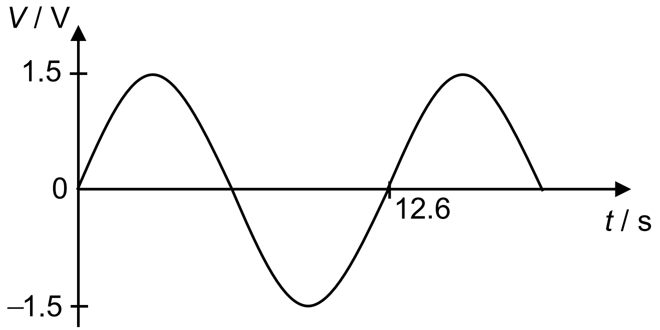
25	The variation of an alternating voltage V with time t is shown in the graph below.	



Which expression best represents V in terms of t ?

	A	$V = 1.5\sin(0.499t)$
	B	$V = 1.5\sin(2.01t)$
	C	$V = 3.0\sin(0.249t)$
	D	$V = 1.5\sin(12.6t)$

Answer: A

$x = x_0 \sin(\omega t)$

$\omega = \frac{2\pi}{T} = \frac{2\pi}{12.6} = 0.499$ rads^{-1}; x_0 = 1.5 V

Therefore, $V = 1.5\sin(0.499t)$.

26 An alternating current with a rectangular waveform as shown in the diagram below flows through a